



CSCI 4620U

Project Part 3a

Conceptual Design, Functionality, and Scenarios (7%)



Due

Sunday, November 14, 11:59pm

Purpose

To give you experience designing (envisioning, specifying, exploring) a solution to your design problem.

Your Tasks

Part 1: Conceptual Design (10%)

Conceptual design is concerned with what the system will do and how it will behave. You will have to design what concepts need to be communicated and how they will be structured, related, and presented. This means deciding which functions your solution will support, how these functions are related, and what information is required to support them. Provide support for your decisions by discussing how the design will deal with: the *problem* you are trying to solve, the results of your *requirements analysis* (from project part 2), the *needs* of your various *stakeholders*. **Highlight any metaphors used in the design.** You can think of this part as a high-level narrative style introduction to your refined (research-informed) project proposal. Note, you should be complete about your concept without worrying you will need to implement everything. We want a full picture of your idea, and usually, only a subset will be prototyped. (Approximate length 1.5 pages)

Part 2: Functionality (15%)

List the specific capabilities of your system. Describe the capabilities in sufficient detail to be clear to someone not on your design team. Note that this may be quite similar to the functionality submitted in part 2; refine and expand the details.

Part 3: Scenarios (25%)

Write at least two *scenarios* describing how primary stakeholders will use your system. At least one scenario must be a *story scenario*. The other may be a story, conceptual, or concrete scenario (label clearly which type you are using). The scenarios should make use of what you have learned from the user needs and requirements analysis, and should illustrate the most important functionality. Your scenarios should use the persona(s) you have created in part 2b. (Approximate length one page)

Part 4: Interaction Design (15%)

Think through, design, and present one example of a key interaction sequence, describing what people will have to do in order to accomplish tasks with your system. Use the *hierarchical task analysis* technique from part 2b. Note that in part 2b your interaction sequence illustrated tasks as users *currently* accomplish them. Here you should describe tasks as they will be accomplished *using your proposed system*.

Part 5: Interface Design (35%)

Carry out an initial design of your solution. This encompasses both *information design* and *interaction design*.

5a Interface Design (20%)

Provide sketches of interface software. The level of detail should be sufficient to convey your design ideas – detailed, artistic drawings are not necessary. Specifically, you should *sketch* the user interface screen(s) (using software or paper/pencil). Describe and illustrate any particularly important or novel interaction elements. If your project involves novel hardware, additional sketches or even physical models to illustrate the size, weight, and form factor would be helpful. These sketches should be inspired by those created by team members in Lab 5. Include a copy of the individual sketches as an appendix.

5b Interaction Design (15%)

Provide sketches of key interaction sequences *in the form of a storyboard*. Note that you may copy/paste electronic sketches or scans from 5a and annotate them to show the interaction sequences. You may also add captions to your storyboard frames. Many sketching techniques and tips are discussed in Module 8. The storyboard should illustrate the particularly important or novel interaction sequences of your design, ideally based on your scenarios.

Appendix

Include your individual contribution report, which lists who completed which parts of your project submission, and notes from any meetings you held. Include individual sketches from Lab 5.

What to Submit

Select one group member as the contact person. This person will submit your assignment as a **single PDF** file using Canvas. Your project report should contain the following:

1. Cover page with group member names **and the name of your private Slack channel you share with the TA and instructor so that we may contact you if needed.**
2. Your conceptual design
3. A functionality list
4. Scenarios
5. A task analysis illustrating the interaction design
6. Interface and interaction design sketches
7. Appendices including *Individual Contribution Report* and *Sketches from Lab 5*

Please name your submission file with your group member names, e.g. “z5-collins-keith-zhao-P3a.pdf”.

Grading

Your assignment will be graded based on the quality and originality of your ideas, your description of the concept, the consistency and clarity of your scenarios, interaction design, and interface design, and how well they are supported by your research results from part 2. Grades may be deducted for assignments which are poorly written or confusing. Check your grammar and spelling.

Academic Integrity

This is a group assignment. While you should work with your group mates, you may not collaborate with other groups, or reuse material from previous offerings of this or similar courses.

In this course, you **MAY NOT** review, copy or otherwise use any graded academic assignments from prior semesters or other sections of this course or similar courses, in written, electronic, or verbal form, used in whole or part, including formatting of any assignment.